
BITCOIN PRICE FORECASTING

Data Science Course Project - Educational Pilot Study

Created in 2024 | English portfolio edition

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Historical analysis. Not financial advice.



Project at a glance

A transparent record of the original course study

Question

- Can historical market and on-chain variables support exploration of a five-day-ahead Bitcoin price forecasting task?
- The project examines methods; it is not a trading system.

Portfolio position

- Educational Data Science course project created in 2024.
- Historical source coverage ends in January 2024.
- No current accuracy or profitability claim is made.

This deck translates and preserves the original analytical storyline while clearly separating historical study outputs from current claims.

Study structure

From historical data to model comparison

Analysis

- Data assembly from Bitcoin, on-chain, macro and market indicators.
- Exploratory views: price path, behavioural-cycle label, seasonality, weekday variation and correlations.
- Feature screening and transformation before modelling.

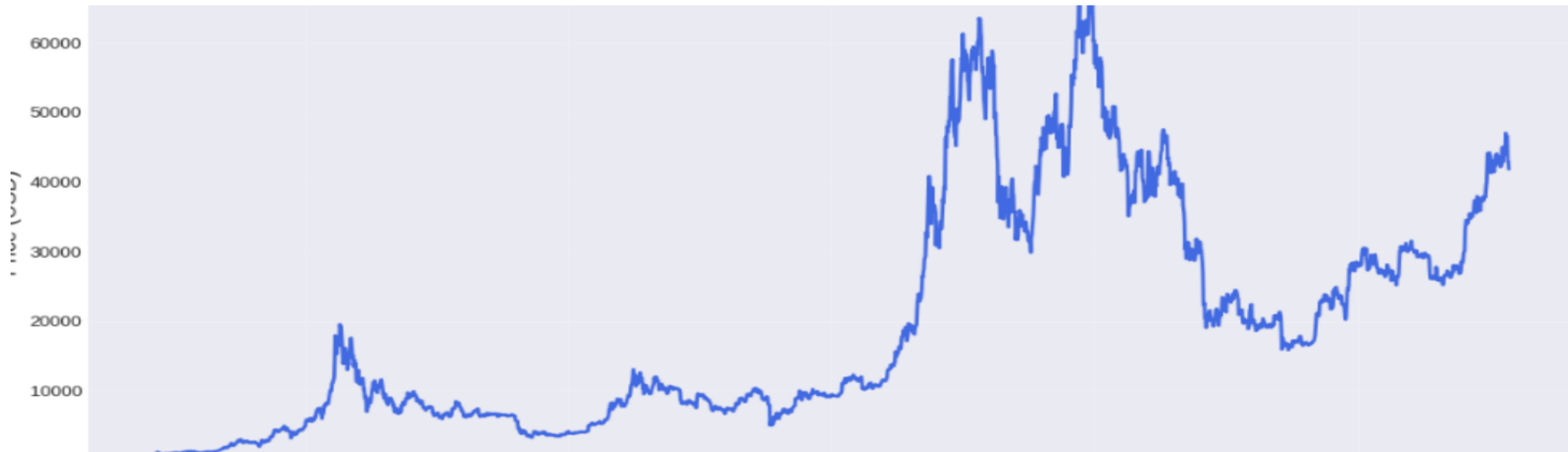
Modelling

- Regression, tree ensembles and neural-network approaches were explored.
- Training and test splits were used in the original course notebook.
- Outputs are historical course-study artefacts and need a time-aware refresh before reuse.

The source notebook and the original charts are preserved locally; this public version deliberately avoids treating past outputs as live results.

Historical Bitcoin price path

Source study visual: long-run price movement



- The original study uses historical price observations as the prediction target.
- The price path is strongly non-stationary and volatile - a key modelling challenge.

Chart reproduced from the original 2024 course presentation. It is historical context, not a market outlook.

Dataset design

31 source datasets and 34 recorded features in the original study

Time coverage

- Bitcoin-related series: 2009 through January 2024.
- Selected market variables: 2015 through January 2024.
- Target: a five-day-ahead Bitcoin price value.

Feature families

- Price, volume, market capitalization and technical indicators.
- On-chain activity: blocks, confirmations, hash rate, transactions and addresses.
- External context: S&P 500, gold, EUR/USD, China energy and coal prices.

The source presentation attributes the original on-chain data category to Blockchain.com; refresh work must record exact retrieval dates and licenses.

Why these variables were explored

The original study combined blockchain activity with external market context

On-chain signals

- Hash rate, difficulty, block and transaction activity can describe network conditions.
- Market capitalization, volume and address activity represent different dimensions of historical activity.

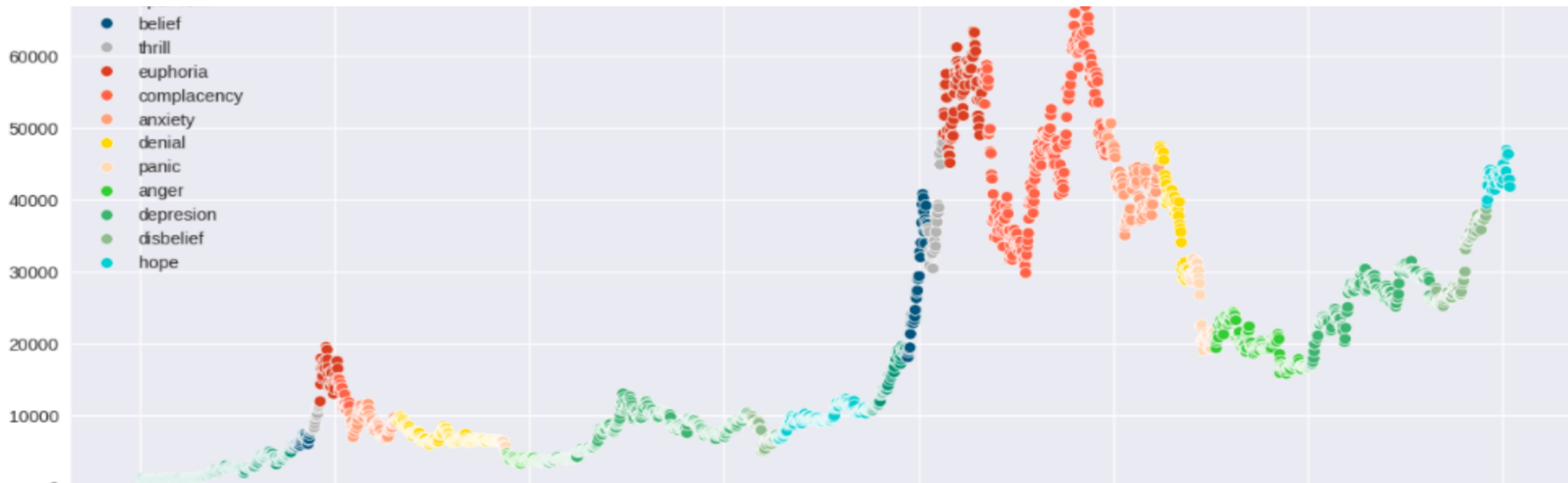
External context

- Gold, EUR/USD and S&P 500 were included as market-context variables.
- Energy and coal prices were explored as possible mining-cost context variables.
- Inclusion does not demonstrate causation.

Variables were selected for exploratory modelling; causal claims would require a separate research design.

Exploratory view: behavioural-cycle labels

Original chart used a labelled market-psychology feature

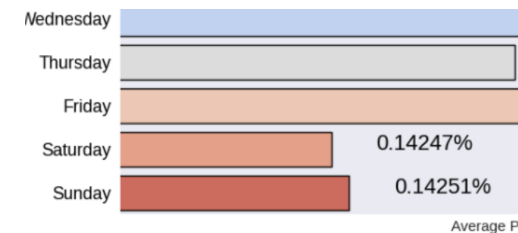
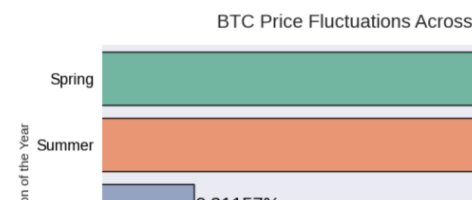
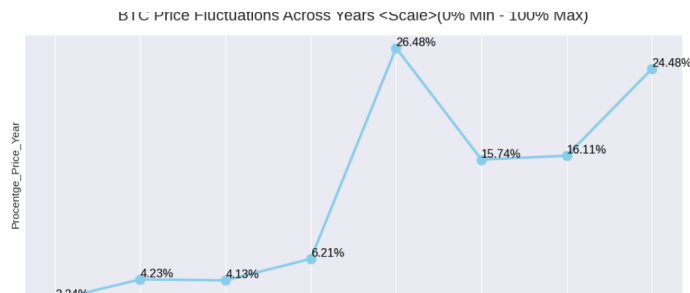


- The source study plotted historical price observations against labelled behavioural-cycle categories.
- This is an exploratory representation, not a validated measure of investor sentiment.

Chart labels are retained from the original analysis. Their predictive value was not independently re-validated for this portfolio edition.

Exploratory views: week, season and weekday

The original analysis inspected recurring patterns before modelling

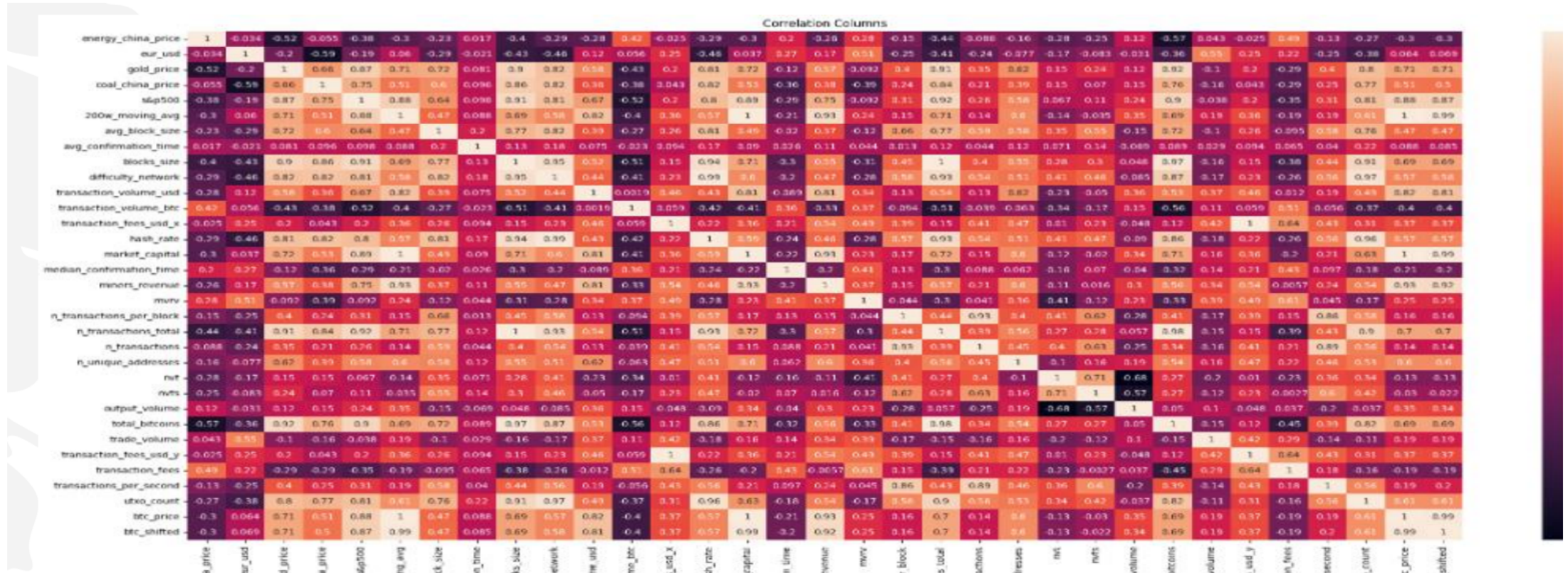


- These views are descriptive summaries of the historical sample.
- Observed variation does not establish stable seasonal or weekday effects and should be retested using walk-forward validation.

The public version retains the evidence but avoids converting descriptive patterns into investment recommendations.

Correlation screening

A wide correlation matrix informed the next feature-review step



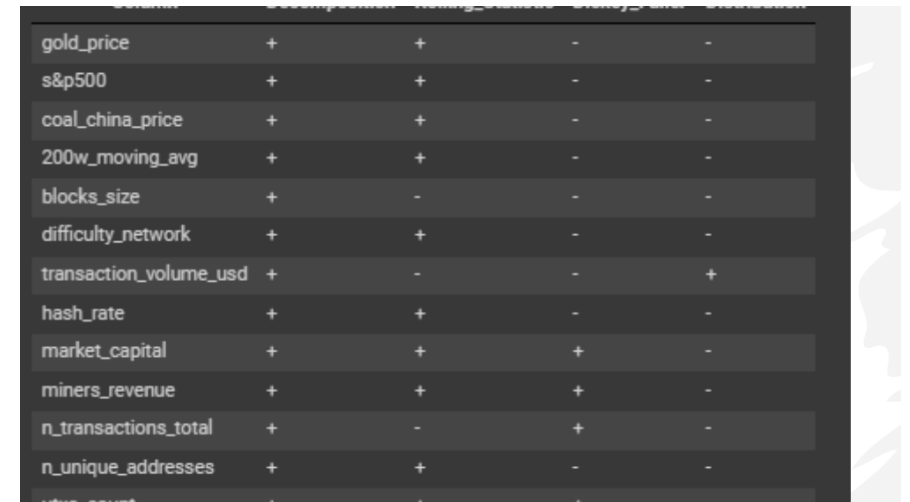
- The correlation review identified groups of highly related features.
- High correlation can increase redundancy and overfitting risk, so it informed subsequent feature screening.

Correlation alone does not select a forecasting feature set; it is one diagnostic in the original exploratory workflow.

Feature screening

The original study combined several diagnostics

- Decomposition: examine trend, seasonal and residual behaviour.
- Rolling statistics: inspect time-varying level and volatility.
- Dickey-Fuller: assess stationarity diagnostics.
- Distribution views: inspect variability and shape.
- The original study notes removal of one feature after screening.



Feature	Decomposition	Rolling Statistics	Dickey-Fuller	Distribution
gold_price	+	+	-	-
s&p500	+	+	-	-
coal_china_price	+	+	-	-
200w_moving_avg	+	+	-	-
blocks_size	+	-	-	-
difficulty_network	+	+	-	-
transaction_volume_usd	+	-	-	+
hash_rate	+	+	-	-
market_capital	+	+	+	-
miners_revenue	+	+	+	-
n_transactions_total	+	-	+	-
n_unique_addresses	+	+	-	-
utxo_count	+	+	+	-

Feature decisions should be fitted inside each training window to avoid information leakage in a refreshed study.

Preprocessing and feature engineering

Original workflow decisions, restated in English

Transformations

- The original study applied logarithmic and seasonal-decomposition transformations to address non-stationary series.
- New calendar-derived fields included daily, seasonal and annual price-change context, seasons and weekdays.

Refresh requirement

- A future version should fit scaling, decomposition and feature selection only on training windows.
- Evaluate with time-aware splits and retain an untouched final holdout period.

This slide describes the original approach. It does not certify that all transformations were leakage-safe.

Model families explored

The source notebook compared multiple regression approaches

Baselines and ensembles

- Linear Regression and K-Nearest Neighbours.
- Decision Tree, Random Forest and Extra Trees.
- AdaBoost and Gradient Boosting.
- XGBoost and LightGBM.

Neural networks

- MLPRegressor.
- Keras Sequential neural-network workflow.
- Standard scaling was applied to inputs for linear and neural-network models in the original study.

Model names record the original course exploration. They are not a recommendation for any deployment or trading use.

Original train/test setup

Historical notebook configuration

Recorded split

- X_train: 2,450 rows x 17 features.
- X_test: 121 rows x 17 features.
- Target vectors: y_train 2,450; y_test 121.

Methodological caution

- The public documentation does not verify a walk-forward evaluation design.
- A refreshed study should compare against naive baselines and report errors in original price units.

These figures are transcribed from the original 2024 course presentation and have not been re-run for this portfolio edition.

Historical model-comparison output

Source-study scores are shown as historical notebook outputs

- The original presentation recorded best pre-tuning R2 values of 0.85 for MLP and 0.86 for Linear Regression, with AdaBoost at 0.80.
- These values are not current, independently validated, or directly comparable without the original evaluation context.

```
MSE = 0.008846475648837839
RMSE = 0.09405570503078396
MAE = 0.07268163640788569
r2 = 0.8054804326559408
```

LinearRegression:

```
MSE = 0.004582210610938365
RMSE = 0.06769202767636943
MAE = 0.05127566923314914
r2 = 0.8627888769554121
```

:

```
MSE = 0.005025683256722897
```

Historical source output only. It must not be read as a production-grade performance claim.

Tuning and regularisation

Original study explored hyperparameters and neural-network penalties

Parameters discussed

- Tree and boosting settings: `n_estimators`, `learning_rate`, `max_depth`, `subsample` and feature sampling.
- Regularisation: L1 and L2 parameters, plus neural-network `alpha` and Keras regularisers.

Recorded historical output

- The original source presentation reports R^2 0.88 for MLP and R^2 0.92 for a Keras Sequential model, with MAE 0.0373 in its transformed evaluation context.
- Those figures require a reproducible refresh before publication as verified performance.

Metric scale and validation design must be made explicit before any score is used for comparison or decision-making.

Observed prediction trajectory

Original source visual: historical target and model-prediction series



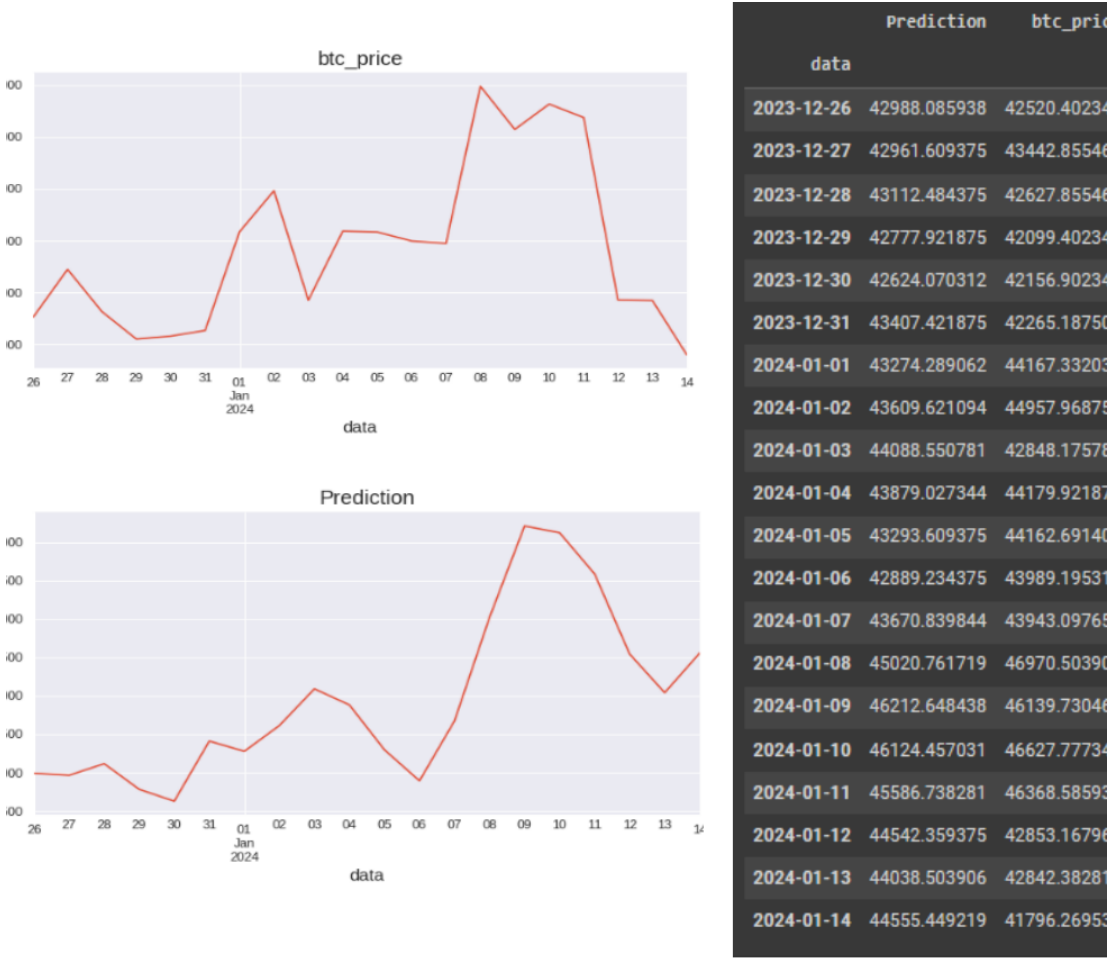
- The course study visual compares historical observations with a model-generated prediction series.
- Visual proximity is not a sufficient evaluation method; an updated study should report error distributions and baseline comparisons.

Chart reproduced from the original course presentation. It is an historical artefact, not a current forecast.

Actual-versus-predicted sample

Source-study comparison for late December 2023 to January 2024

- The original presentation shows two time-series views plus a table of predicted and recorded BTC prices.
- It reports daily deviations ranging from 1.08% to 6.19% for examples shown in that presentation.
- The figures are historical course artefacts and were not independently re-computed here.



A refreshed evaluation should report errors on the original price scale and identify the exact held-out period.

Responsible interpretation

What this project demonstrates - and what it does not

It demonstrates

- A structured 2024 course exploration of data assembly, exploratory analysis, feature screening and model comparison.
- An understandable visual trail from historical indicators to prediction experiments.

It does not demonstrate

- A profitable trading strategy.
- A live or production-ready forecasting system.
- Current prediction accuracy or investment suitability.
- Financial advice, investment advice or a recommendation to buy, sell or hold any asset.

Bitcoin and other crypto-assets are volatile. This is an educational Data Science pilot study, not financial advice.